

CEMAL “BERK” BIRYOL

SENIOR SOFTWARE ENGINEER

CONTACT

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EDUCATION

Ph.D. Geophysics & Seismology,
University of Arizona, 2009.

M.Sc. Geological Engineering, Middle
East Technical University, 2004.

B.Sc. Geological Engineering, Middle
East Technical University, 2002.

**Full-Stack Web Development
Certificate**, University of North Carolina,
2018.

SKILLS

- **Languages & Frameworks:** VB.NET, C#, JavaScript, TypeScript, Python, SQL, .NET, React, Node.js.
- **Web & API:** RESTful APIs, SOAP APIs, GraphQL (basic), HTML5, CSS3.
- **Databases:** MySQL, SQL Server, PostgreSQL, SQLite; stored procedures, views.
- **DevOps & Tools:** Docker, GitHub Actions, Jenkins, Git, CI/CD pipelines.
- **Cloud Platforms:** AWS (S3, EC2, RDS, Elastic Beanstalk, Fargate, CloudWatch), limited GCP
- **Testing & QA:** Unit testing, integration testing, PyTest, Selenium (basic)
- **Project Management:** Agile/Scrum, Jira, cross-functional collaboration
- **Other:** Linux/Ubuntu, SAS (limited), public sector consulting, technical mentoring

PROFESSIONAL OVERVIEW

Experienced Full-Stack Software Engineer with a Ph.D. in Geophysics and a proven track record building scalable, data-driven applications for both government and private-sector clients. Skilled in .NET (VB.NET, C#), React, Node.js, SQL optimization, and AWS. Strong background in API development, Docker-based DevOps, and technical leadership. Cleared to work on EPA and NIH projects through ICF consulting roles.

WORK EXPERIENCE

ICF – Durham, NC

Senior Software Engineer | 2021 – 2025

- Built scalable full-stack solutions for public health and environmental programs at NIH and EPA, leveraging AWS, Docker, and SQL-based backends.
- Developed and maintained RESTful APIs and internal tools to process and analyze large air quality datasets, including limited work with SAS.
- Improved performance of SQLAlchemy-based Python applications by optimizing queries.
- Led debugging efforts for production systems, resolving issues with data load times and container deployments.
- Contributed to CI/CD pipelines with GitHub Actions and mentored team members.
- Held Public Trust clearance as a badged contractor.

Regency360 – Raleigh, NC

Software Engineer | 2018 – 2021

- Developed customer-facing SOAP API integrations for order and product management using VB.NET, C#, and .NET Framework.
- Created seamless e-commerce workflows for card payments and promotional product orders through vendor-defined WSDL templates.
- Built and maintained backend systems with Microsoft SQL Server and MySQL, including complex stored procedures and views.
- Designed live dashboards for financial and order data visualization, integrating real-time metrics for sales teams.
- Supported internal platform services and business automation through full-stack development.